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PRACTICAL 10

Code:

```
<> prr10.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8">
5    <title>Practical 10 - JSON Handling with Fetch and jQuery</title>
6    <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
7    <style>
8      body {
9        font-family: Arial, sans-serif;
10       text-align: center;
11       background: #f7f9fc;
12       margin: 20px;
13     }
14     table {
15       width: 80%;
16       margin: 20px auto;
17       border-collapse: collapse;
18       background-color: #fff;
19       box-shadow: 0 0 8px rgba(0,0,0,0.1);
20     }
21     th, td {
22       border: 1px solid #ccc;
23       padding: 10px;
24     }
25     th {
26       background-color: #007bff;
27       color: white;
28     }
29     tr.highlight {
30       background-color: #bdf7bd !important; /* highlight toppers */
31       font-weight: bold;
32     }
33   </style>
34 </head>
35 <body>
36
37   <h2> Student Records (Using Fetch and jQuery AJAX)</h2>
38
39   <h3> Data Loaded Using Fetch()</h3>
40   <table id="fetchTable">
41     <thead></thead>
42     <tbody></tbody>
43   </table>
44
45   <h3> Data Loaded Using $.getJSON()</h3>
46   <table id="jqueryTable">
47     <thead></thead>
48     <tbody></tbody>
49   </table>
50
```

```
{ } data.json > ...
1  [
2    { "id": 1, "name": "Alice", "marks": 85, "grade": "A" },
3    { "id": 2, "name": "Bob", "marks": 92, "grade": "A+" },
4    { "id": 3, "name": "Charlie", "marks": 78, "grade": "B" },
5    { "id": 4, "name": "David", "marks": 95, "grade": "A+" },
6    { "id": 5, "name": "Esha", "marks": 67, "grade": "C" }
7  ]
8
```

```

JS script.js > ...
1 // ----- Fetch API Example -----
2 fetch('data.json')
3   .then(response => response.json())
4   .then(data => {
5     // Simulate delay
6     setTimeout(() => {
7       let thead = document.querySelector('#fetchTable thead');
8       let tbody = document.querySelector('#fetchTable tbody');
9
10      thead.innerHTML = "<tr><th>ID</th><th>Name</th><th>Marks</th><th>Grade</th></tr>";
11      tbody.innerHTML = "";
12
13      data.forEach(student => {
14        const row = document.createElement('tr');
15        if (student.marks > 90) row.classList.add('highlight');
16        row.innerHTML = `
17          <td>${student.id}</td>
18          <td>${student.name}</td>
19          <td>${student.marks}</td>
20          <td>${student.grade}</td>
21        `;
22        tbody.appendChild(row);
23      });
24    }, 1500); // 1.5 sec delay to observe async loading
25  })
26  .catch(error => console.error('Fetch Error:', error));
27
28 // ----- jQuery AJAX Example -----
29 $.getJSON('data.json', function(data) {
30   let table = $('#jqueryTable');
31   table.find('thead').html("<tr><th>ID</th><th>Name</th><th>Marks</th><th>Grade</th></tr>");
32   let tbody = table.find('tbody');
33   tbody.empty();
34
35   $.each(data, function(index, student) {
36     let row = `<tr class='${student.marks > 90 ? "highlight" : ""}'>
37       <td>${student.id}</td>
38       <td>${student.name}</td>
39       <td>${student.marks}</td>
40       <td>${student.grade}</td>
41     </tr>`;
42     tbody.append(row);
43   });
44   }).fail(function() {
45     console.error("Error loading JSON data with jQuery.");
46   });
47

```

Output :

Student Records (Using Fetch and jQuery AJAX)

Data Loaded Using Fetch()

ID	Name	Marks	Grade
1	Alice	85	A
2	Bob	92	A+
3	Charlie	78	B
4	David	95	A+
5	Esha	67	C

Data Loaded Using \$.getJSON()

ID	Name	Marks	Grade
1	Alice	85	A
2	Bob	92	A+
3	Charlie	78	B
4	David	95	A+
5	Esha	67	C